Model Question Paper Set - 1 with effect from 2022(CBCS Scheme)

USN | | | | | | | | | | |

Fourth Semester B.E Degree Examination

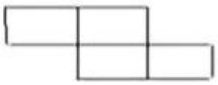
DISCRETE MATHEMATICAL STRUCTURES (BCS405A)

TIME: 03Hours **Max.Marks:100**

Note:

1. Answer any **FIVE** full questions, choosing at least **ONE** question from each **MODULE**
2. M: Marks, L: RBT levels, C: Course outcomes.

		Module – 1	M	L	C
Q.1	a	Show that the compound proposition $[(p \leftrightarrow q) \wedge (q \leftrightarrow r) \wedge (r \leftrightarrow p)] \Leftrightarrow [(p \rightarrow q) \wedge (q \rightarrow r) \wedge (r \rightarrow p)]$ for primitive statements p, q, r is logically equivalence.	6	L2	CO1
	b	Establish the validity of the following argument using the Rules of Inference: $\{p \wedge (p \rightarrow q) \wedge (s \vee r) \wedge (r \rightarrow \sim q)\} \rightarrow (s \vee t)$.	7	L2	CO1
	c	For the universe of all integers, let p(x), q(x), r(x), s(x) and t(x) denote the following open statements: p(x): x>0, q(x): x is even, r(x): x is a perfect square, s(x): x is divisible by 3, t(x): x is divisible by 7. Write the following statements in symbolic form: i) At least one integer is even. ii) There exists a positive integer that is even. iii) If x is even, then x is not divisible by 3. iv) No even integer is divisible by 7. v) There exists even integer divisible by 3.	7	L1	CO1
		OR			
Q.2	a	Define a tautology. Prove that, for any propositions p, q, r the compound propositions, $\{(p \rightarrow q) \wedge (q \rightarrow r)\} \rightarrow \{(p \rightarrow r)\}$ is tautology.	6	L2	CO1
	b	Prove the following using laws of logic: $p \rightarrow (q \rightarrow r) \Leftrightarrow (p \wedge q) \rightarrow r$.	7	L2	CO1
	c	Give i) direct proof ii) indirect proof iii) proof by contradiction for the following statement: "if n is an odd integer then n+9 is an even integer".	7	L3	CO1
		Module – 2			
Q.3	a	Prove that $1^2 + 3^2 + 5^2 + \cdots + (2n-1)^2 = \frac{n(2n+1)(2n-1)}{3}$ by Mathematical Induction.	6	L2	CO2
	b	Let $a_0 = 1, a_1 = 2, a_2 = 3$ and $a_n = a_{n-1} + a_{n-2} + a_{n-3}$ for $n \geq 3$. Prove that $a_n \leq 3^n \ \forall \ n \in z^+$.	7	L2	CO2
	c	Find the number of ways of arrangement of the letters of the word 'TALLAHASSEE' which have no adjacent A's.	7	L2	CO2
		OR			
Q.4	a	Determine the coefficient of xyz^2 in the expansion of $(2x - y - z)^4$.	6	L2	CO2
	b	In how many ways one can distribute 8 identical marbles in 4 distinct containers so that i) no container is empty ii) the fourth container has an odd number of marbles in it.	7	L2	CO2
	c	How many positive integers n can we form using the digits 3,4,4,5,5,6,7 if we want n to exceed 5,000,000?	7	L2	CO2
		Module – 3			
Q.5	a	Let f: R → R be defined by, $f(x) = \begin{cases} 3x - 5 \ if \ x > 0 \\ 1 - 3x \ if \ x \leq 0 \end{cases}$. Find $f^{-1}(0), f^{-1}(1), f^{-1}(-1), f^{-1}(3), f^{-1}(6), f^{-1}([-6,5])$ and $f^{-1}([-5,5])$	6	L2	CO3

	b	State Pigeon hole principle. Prove that in any set of 29 persons; at least 5 persons have been born on the same day of the week.	7	L2	CO3
	c	Let A={1,2,3,4,6} and 'R' be a relation on 'A' defined by aRb if and only if "a is multiple of b" represent the relation 'R' as a matrix, draw the diagraph and relation R.	7	L2	CO3
		OR			
Q.6	a	If $f:A\to B$, $g:B\to C$, $h:C\to D$ are three functions, then Prove that $h\circ(g\circ f)=(h\circ g)\circ f$.	6	L2	CO3
	b	Show that if n+1 numbers are chosen from 1 to 2n then at least one pair add to 2n+1.	7	L2	CO3
	c	Draw the Hasse diagram representing the positive divisors of 72.	7	L2	CO3
		Module – 4			
Q.7	a	In how many ways the 26 letters of English alphabet are permuted so that none of the pattern's CAR, DOG, PUN or BYTE occurs?	6	L2	CO4
	b	Define Derangement. In how many ways can each of 10 people select a left glove and a right glove out of a total of 10 pairs of gloves so that no person selects a matching pair of gloves?	7	L2	CO4
	c	Solve the recurrence relation: $C_n = 3C_{n-1} - 2C_{n-2}, for\ n \geq 2$, given $C_1 = 5, C_2 = 3$.	7	L3	CO4
		OR			
Q.8	a	In how many ways one can arrange the letters of the word **CORRESPONDENTS** so that there are i) exactly 2 pairs of consecutive identical letters? ii) at least 3 pairs of consecutive identical letters? iii) no pair of consecutive identical letters?	6	L2	CO4
	b	Find the rook polynomial for the chess board as shown in the figure 	7	L2	CO4
	c	Solve the recurrence relation $a_{n+2} - 3a_{n+1} + 2a_n = 0, a_0 = 1, a_1 = 6$.	7	L3	CO4
		Module – 5			
Q.9	a	If H, K are subgroups of a group G, prove that $H \cap K$ is also a subgroup of G. Is $H \cup K$ a subgroup of G.	6	L2	CO5
	b	Define Klein 4 group. Verify $A = \{1, 3, 5, 7\}$ is a klein 4 group.	7	L2	CO5
	c	State and prove Lagrange's Theorem.	7	L2	CO5
		OR			
Q.10	a	Show that i) the identity of G is unique. ii) the inverse of each element of G is Unique.	6	L3	CO5
	b	Show that $(A, \cdot)$ is an abelian group where $A = \{a \in Q \mid a \neq -1\}$ and for any $a, b \in A, a.b = a + b + ab$.	7	L3	CO5
	c	Let $G = S_4$, for $\alpha = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix}$, find the subgroup $H = \langle\alpha\rangle$. Determine the left cosets of H in G.	7	L3	CO5

Model Question Paper Set - 2 with effect from 2022(CBCS Scheme)

USN | | | | | | | | | | |

Fourth Semester B.E Degree Examination

DISCRETE MATHEMATICAL STRUCTURES (BCS405A)

TIME:03Hours **Max.Marks:100**

Note:

1. Answer any **FIVE** full questions, choosing at least **ONE** question from each **MODULE**
2. M: Marks, L: RBT levels, C: Course outcomes.

		Module - 1	M	L	C
Q.1	a	Define tautology. Show that $[(p \vee q) \wedge (p \rightarrow r) \wedge (q \rightarrow r)] \rightarrow r$ is a tautology by constructing the truth table.	6	L1	CO1
	b	Prove the following using the laws of logic $[\neg p \wedge (\neg q \wedge r)] \vee [(q \wedge r) \vee (p \wedge r)] \Leftrightarrow r$.	7	L2	CO1
	c	For any two odd integers m and n, show that (i) m+n is even (ii) mn is odd.	7	L2	CO1
		OR			
Q.2	a	Define i) an open statement ii) Quantifiers	6	L2	CO1
	b	Write the following argument in symbolic form and then establish the validity If A gets the Supervisor's position and works hard, then he will get a raise. If he gets a raise, then he will buy a car. He has not purchased a car. Therefore he did not get the Supervisor's position or he did not work hard.	7	L1	CO1
	c	For the following statements, the universe comprises all non-zero integers. Determine the truth value of each statement. a) $\exists x \exists y [xy = 1]$ b) $\exists x \forall y [xy = 1]$ c) $\forall x \exists y [xy = 1]$ d) $\exists x \exists y [(2x + y = 5) \wedge (x - 3y = -8)]$ e) $\exists x \exists y [(3x - y = 7) \wedge (2x + 4y = 3)]$	7	L2	CO1
		Module – 2			
Q.3	a	Define the well ordering principle. By Mathematical Induction, Prove that $(n!) \geq 2n-1$ for all integers $n \geq 1$.	6	L2	CO2
	b	Prove that every positive integer n≥24 can be written as a sum of 5's and/or 7's.	7	L3	CO2
	c	How many positive integers n , can we form using the digits 3,4,4,5,5,6,7, if we want n to exceed 5,000,000.	7	L1	CO2
		OR			
Q.4	a	By Mathematical Induction Prove that $1.3 + 2.4 + \cdots \ldots \ldots + n(n+2) = \frac{n(n+1)(2n+7)}{6}$.	6	L1	CO2
	b	Find the number of permutations of the letters of the word MASSASAUGA. In how many of these all four $A's$ are together? How many of them begin with S?	7	L2	CO2
	c	i) Obtain the Coefficient of a^5b^2 in the expansion of $(2a-3b)^7$ ii) Using the Binomial theorem find the coefficient of x^5y^2 in	7	L1	CO2

		Module – 3			
Q.5	a	State Pigeon –hole principle. Prove that if any number from1 to 8 are chosen then two of them will have their sum as 9.	6	L1	CO3
	b	Let $f: R \to R$ be defined by ,$f(x) = \begin{cases} 3x-5, & if\ x > 0 \\ 1-3x, & if x \leq 0 \end{cases}$ $find, f^{-1}([-6,5])\ and\ f^{-1}([-5,5])$.	7	L1	CO3
	c	Let $A = B = C = R$, and $f: A \to B$ and $g: B \to C$ be defined by $f(a) = 2a + 1, \quad g(b) = \frac{1}{3}b, \forall a \in A, \forall b \in B.$ Compute gof and show that gof is invertible. What is $(g\ o\ f)^{-1}$?	7	L2	CO3
		OR			
Q.6	a	Let f and g be functions from R to R defined by $f(x) = ax + b$ and $g(x) = 1 - x + x^2$. If $(g\ o\ f)(x) = 9x^2 - 9x + 3$ determine a and b.	6	L3	CO3
	b	Draw the Hasse (POSET) diagram which represents positive divisors of 36.	7	L2	CO3
	c	Let $A = \{1,2,3,4,6\}$ and R be a relation on A defined by aRb if and only if "a is a multiple of b". Write down the relation R, relation matrix $M(R)$ And draw its diagraph. List out its in degree and out degree.	7	L2	CO3
		Module – 4			
Q.7	a	Determine the number of positive integers n such that $1 \leq n \leq 100$ and n is not divisible by 2, 3, or 5	6	L2	CO4
	b	In how many ways can the 26 letters of the English alphabet be permuted so that none of the patterns CAR, DOG, PUN or BYTE occurs?	7	L2	CO4
	c	Solve the recurrence relation $a_n = na_{n-1}$ where n ≥ 1and $a_0 = 1$.	7	L2	CO4
		OR			
Q.8	a	In how many ways 5 number of a's, 4 number of b's and 3 number of c's can be arranged so that all the identical letters are not in a single block?	6	L3	CO4
	b	Five teachers T_1, T_2, T_3, T_4, T_5 are to be made class teachers for five classes, C_1, C_2, C_3, C_4, C_5, one teacher for each class. $T_1$and T_2 do not wish to become the class teachers for C_1or C_2, T_3 and T_4 for C_4 or C_5 , and $T_5$for C_3 or C_4 or C_5 .In how many ways can the teachers be assigned the work (without displeasing any teacher	7	L2	CO4
	c	Solve the recurrence relation $F_{n+2} = F_{n+1} + F_n$where n ≥ 0 and $F_0 = 0, F_1 = 1$.	7	L2	CO4
		Module – 5			
Q.9	a	Define group. Show that fourth roots of unity is an abelian group.	6	L2	CO5
	b	If G be a set of all non-zero real numbers and let a* b = ab/2 then show that (G,*) is an abelian group.	7	L2	CO5
	c	Define Klein 4-group. And if A = { e,a,b,c} then show that this is a Klein -4 group	7	L1	CO5
		OR			
Q.10	a	Define Cyclic group and show that (G,8) whose multiplication table is as given below is Cyclic	6	L2	CO5

*	a	b	c	d	e	f
a	a	b	c	d	e	f
b	b	c	d	e	f	a
c	c	d	e	f	a	b
d	d	e	f	a	b	c
e	e	f	a	b	c	d
f	f	a	b	c	d	e

b	State and prove Lagrange's theorem	7	L1	CO5
c	If G be a group with subgroup H and K. If $\|G\| = 660$ and $\|K\| = 66$ and K C H C G and find the possible value for $\|H\|$	7	L2	CO5

CBCS SCHEME

7/19

BCS405A

USN

Fourth Semester B.E./B.Tech. Degree Examination, June/July 2024
Discrete Mathematical Structures

Time: 3 hrs. Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.

		Module – 1	M	L	C
Q.1	a.	Define tautology. Prove that for any propositions p, q, r the compound proposition. $[(p \wedge \neg q) \rightarrow r] \rightarrow [p \rightarrow (q \vee r)]$ is a tautology	06	L2	CO1
	b.	Test whether the following is a valid argument: If Ram studies then he will pass 12th. If Ram passes 12th then his father gifts him a bike. If Ram doesn't play video game then he will pass 12th. Ram did not get a bike. ∴ Ram played video game.	07	L3	CO1
	c.	Give direct proofs of the statements: i) If k and *l* are odd then k + *l* is even. ii) If k and *l* are odd then k*l* is odd.	07	L2	CO1
		OR			
Q.2	a.	Define (i) Proposition (ii) Open statement (iii) Quantifiers	06	L2	CO1
	b.	Using the laws of logic, prove the following logical equivalence: $[(\neg p \vee \neg q) \wedge (F_0 \vee p) \wedge p] \Leftrightarrow p \wedge \neg q$	07	L2	CO1
	c.	Write the following statement in symbolic form and find its negation: "If all triangles are right angled then no triangle is equilateral".	07	L2	CO1
		Module – 2			
Q.3	a.	Prove by using mathematical induction. $1^2 + 2^2 + \ldots\ldots + n^2 = \frac{n(n+1)(2n+1)}{6}$	06	L2	CO1
	b.	How many words can be made with or without meaning from the letters of the word "STATISTICS"? In how many of these a and c are adjacent? In how many vowels are together?	07	L3	CO2
	c.	Find the coefficient of x^3y^8 in the expansion of $(2x - y)^{11}$.	07	L2	CO2
		OR			
Q.4	a.	Obtain the recursive definition for the sequence in each of the following cases: (i) $a_n = 5n$ (ii) $a_n = 3n + 7$ (iii) $a_n = n^2$ (iv) $a_n = 2 - (-1)^n$	06	L2	CO2
	b.	A woman has 11 close relations and wishes to invite 5 of them to dinner. In how many ways can she invite them if (i) there is no restriction on her choice. (ii) 2 persons will not attend separately (iii) 2 persons will not attend together.	07	L3	CO2
	c.	In how many ways can we distribute 7 apples and 5 oranges among 3 children such that each child gets atleast one apple and one orange?	07	L3	CO2

CBCS SCHEME

USN | | | | | | | | | |

BCS405A

Fourth Semester B.E./B.Tech. Degree Examination, June/July 2024
Discrete Mathematical Structures

Time: 3 hrs. Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.

		Module – 1	M	L	C
Q.1	a.	Define tautology. Prove that for any propositions p, q, r the compound proposition. $[(p \wedge \neg q) \rightarrow r] \rightarrow [p \rightarrow (q \vee r)]$ is a tautology	06	L2	CO1
	b.	Test whether the following is a valid argument: If Ram studies then he will pass 12th. If Ram passes 12th then his father gifts him a bike. If Ram doesn't play video game then he will pass 12th. Ram did not get a bike. ∴ Ram played video game.	07	L3	CO1
	c.	Give direct proofs of the statements: i) If k and *l* are odd then k + *l* is even. ii) If k and *l* are odd then k*l* is odd.	07	L2	CO1
		OR			
Q.2	a.	Define (i) Proposition (ii) Open statement (iii) Quantifiers	06	L2	CO1
	b.	Using the laws of logic, prove the following logical equivalence: $[(\neg p \vee \neg q) \wedge (F_0 \vee p) \wedge p] \Leftrightarrow p \wedge \neg q$	07	L2	CO1
	c.	Write the following statement in symbolic form and find its negation: "If all triangles are right angled then no triangle is equilateral".	07	L2	CO1
		Module – 2			
Q.3	a.	Prove by using mathematical induction. $1^2 + 2^2 + \ldots\ldots + n^2 = \frac{n(n+1)(2n+1)}{6}$	06	L2	CO1
	b.	How many words can be made with or without meaning from the letters of the word "STATISTICS"? In how many of these a and c are adjacent? In how many vowels are together?	07	L3	CO2
	c.	Find the coefficient of x^3y^8 in the expansion of $(2x - y)^{11}$.	07	L2	CO2
		OR			
Q.4	a.	Obtain the recursive definition for the sequence in each of the following cases: (i) $a_n = 5n$ (ii) $a_n = 3n + 7$ (iii) $a_n = n^2$ (iv) $a_n = 2 - (-1)^n$	06	L2	CO2
	b.	A woman has 11 close relations and wishes to invite 5 of them to dinner. In how many ways can she invite them if (i) there is no restriction on her choice. (ii) 2 persons will not attend separately (iii) 2 persons will not attend together.	07	L3	CO2
	c.	In how many ways can we distribute 7 apples and 5 oranges among 3 children such that each child gets atleast one apple and one orange?	07	L3	CO2

		Module – 3			
Q.5	a.	State pigeon hole principle. Using pigeon hole principle find the minimum number of persons chosen so that atleast 5 of them will have their birthday in the same month.	06	L3	CO3
	b.	Let A = {a, b, c, d} and B = {1, 2, 3, 4, 5}. Find the number of 1-1 functions and onto functions from (i) A to B (ii) B to A	07	L2	CO3
	c.	Let A = {1, 2, 3, 4, 5}. Define a relation R on A × A by (x_1, y_1) R (x_2, y_2) iff $x_1 + y_1 = x_2 + y_2$. (i) Verify that R is an equivalence relation (ii) Determine the equivalence class of [(2, 4)]	07	L2	CO3
		OR			
Q.6	a.	Consider the functions f and g from R to R defined by f(x) = 2x + 5 and g(x) = ½ (x – 5). Prove that g is inverse of f.	06	L2	CO3
	b.	Let A = {1, 2, 3, 4} and R be the relation on A defined by xRy if and only if x < y. Write down R as a set of ordered pairs. Write the relation matrix and draw the digraph. List out the in degrees and out degrees of every vertex.	07	L2	CO3
	c.	Let A = {1, 2, 3, 6, 9, 12, 18} and define R on A by xRy iff 'x divides y'. Prove that (A, R) is a POSET. Draw the Hasse diagram for (A, R).	07	L2	CO3
		Module – 4			
Q.7	a.	How many integers between 1 and 300 (inclusive) are divisible by (i) atleast one of 5, 6 or 8. (ii) None of 5, 6 and 8.	06	L3	CO4
	b.	At a restaurant 10 men handover their umbrellas to the receptionist, In how many ways can their umbrellas be returned so that (i) no man receives his own umbrella. (ii) atleast one gets his own umbrella. (iii) atleast two gets their own umbrellas.	07	L3	CO4
	c.	The number of virus affected files in a system is 1000 (to start with) and this increases by 250% every 2 hours. Use a recurrence relation to determine the number of virus affected files in the system after 12 hours.	07	L3	CO4
		OR			
Q.8	a.	In how many ways one can arrange the letters of the word "CORRESPONDENTS" so that there are (i) no pair (ii) atleast 2 pairs of consecutive identical letters.	06	L3	CO4
	b.	4 persons P_1 , P_2 , P_3 , P_4 who arrive late for a dinner party find that only one chair at each of five tables T_1 , T_2 , T_3 , T_4 and T_5 is vacant. P_1 will not sit at T_1 or T_2. P_2 will not sit at T_2. P_3 will not sit at T_3 or T_4. P_4 will not sit at T_4 or T_5. Find the number of ways they can occupy the vacant chairs.	07	L3	CO4
	c.	Solve the recurrence relation $a_n - 6a_{n-1} + 9a_{n-2} = 0$ for $n \geq 2$ with $a_0 = 5$, $a_1 = 12$.	07	L2	CO4
		Module – 5			
Q.9	a.	If * is an operation on Z defined by $xy = x + y + 1$, prove that (Z, *) is an abelian group.	06	L2	CO5
	b.	Explain Klein-4 group with example.	07	L2	CO5
	c.	State and prove Lagrange's theorem.	07	L2	CO5
		OR			
Q.10	a.	Prove that intersection of two subgroups of a group G is also a subgroup of G.	06	L2	CO5
	b.	Prove that $(Z_4 , +)$ is a cyclic group. Find all its generators.	07	L2	CO5
	c.	Let $G = S_4$ for $\alpha = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix}$ Find the subgroup $H = \langle \alpha \rangle$ determine the left cosets of H in G.	07	L3	CO5

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Fourth Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025
Discrete Mathematical Structures

Time: 3 hrs. Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.

		Module – 1	M	L	C
Q.1	a.	Define a tautology. Prove that for any propositions p, q, r the compound propositions $\{(p \rightarrow q) \wedge (q \rightarrow r)\} \rightarrow (p \rightarrow r)$ is tautology.	06	L2	CO1
	b.	Establish the validity of the following argument using the rules of inference: $\{p \wedge (p \rightarrow q) \wedge (s \vee r) \wedge (r \rightarrow \sim q)\} \rightarrow (s \vee t)$	07	L2	CO1
	c.	For any two odd integers m and n, show that: (i) m + n is even (ii) mn is odd	07	L2	CO1
		OR			
Q.2	a.	Show that the compound proposition $[(p \vee q) \rightarrow r] \Leftrightarrow [(p \rightarrow r) \wedge (q \rightarrow r)]$ for primitive statements p, q, r is logically equivalent.	06	L2	CO1
	b.	Prove the following using law of logic: $p \rightarrow (q \rightarrow r) \Leftrightarrow (p \wedge q) \rightarrow r$	07	L2	CO1
	c.	Determine the truth value of each of the following quantified statements, the universe being the set of all non-zero integers: (i) $\exists x, \exists y, [xy = 1]$ (ii) $\exists x, \forall y, [xy = 1]$ (iii) $\forall x, \exists y, [xy = 1]$ (iv) $\exists x, \exists y, [(2x + y = 5) \wedge (x - 3y = -8)]$ (v) $\exists x, \exists y, [(3x - y = 17) \wedge (2x + 4y = 3)]$	07	L3	CO1
		Module – 2			
Q.3	a.	Prove that for each $n \in z^+$, $1^2 + 2^2 + 3^2 + \ldots + n^2 = \frac{n(n+1)(2n+1)}{6}$.	06	L2	CO2
	b.	Let $a_0 = 1$, $a_1 = 2$, $a_2 = 3$ and $a_n = a_{n-1} + a_{n-2} + a_{n-3}$ for $n \geq 3$, prove that $a_n \leq 3^n \ \forall \ n \in z^+$.	07	L2	CO2
	c.	How many positive integers n can be we form using the digits 3, 4, 4, 5, 5, 6, 7 if we want n to exceed 5,000,000?	07	L3	CO2
		OR			
Q.4	a.	By mathematical induction prove that $1.3 + 2.4 + \ldots + n(n+2) = \frac{n(n+1)(2n+7)}{6}$	06	L2	CO2
	b.	Find the number of permutations of the letters of the word ENGINEERING such that: (i) All the E's are together (ii) Arrangement begin with N (iii) All the vowels are adjacent.	07	L3	CO2
	c.	Find the coefficient of $a^2b^3c^2d^5$ in the expansion of $(a + 2b - 3c + 2d + 5)^{16}$.	07	L3	CO2
		Module – 3			
Q.5	a.	State pigeon hole principle. Prove that if 30 dictionaries in a library contain a total of 61,327 pages then atleast one of the dictionaries must have atleast 2045 pages.	06	L3	CO3
	b.	Let $f : R \rightarrow R$ be defined by $f(x) = \begin{cases} 3x - 5 & \text{if } x > 0 \\ 1 - 3x & \text{if } x \leq 0 \end{cases}$. Find $f^{-1}(0)$, $f^{-1}(1)$, $f^{-1}(-1)$, $f^{-1}(3)$, $f^{-1}(-6)$, $f^{-1}([-6, 5])$ and $f^{-1}([-5, 5])$	07	L2	CO3
	c.	Draw the Hasse diagram representing the positive divisor of 36.	07	L3	CO3

		OR			
Q.6	**a.**	Let A = {1, 2, 3, 4} and B = {1, 2, 3, 4, 5, 6}, (i) How many functions are there from A to B? (ii) How many of these are one to one? (iii) How many functions are there from B to A? (iv) How many of these are onto?	**06**	**L2**	**CO3**
	b.	Let f and g be functions from R to R defined by $f(x) = ax + b$ and $g(x) = 1 - x + x^2$. If $(g \circ f)(x) = 9x^2 - 9x + 3$, determine a and b.	**07**	**L2**	**CO3**
	c.	Let A = {1, 2, 3, 4, 6} and R be a relation on A defined by aRb if and only if "a is multiple of b". Write down the relation R, relation matrix M(R) and draw the digraph. List out in degree and out degree.	**07**	**L3**	**CO3**
		Module – 4			
Q.7	**a.**	In how many ways 5 number of a's, 4 number of b's and 3 number of c's can be arranged so that all the identical letters are not in a single block?	**06**	**L3**	**CO4**
	b.	Determine the number of positive integers n such that $1 \le n \le 100$ and n is not divisible by 2, 3, or 5.	**07**	**L3**	**CO4**
	c.	Solve the recurrence relation $a_{n+2} - 3a_{n+1} + 2a_n = 0$, $a_0 = 1$, $a_1 = 6$.	**07**	**L2**	**CO4**
		OR			
Q.8	**a.**	In how many ways can the 26 letters of the English alphabet be permuted so that none of the patterns CAR, DOG, PUN or BYTE occurs?	**06**	**L3**	**CO4**
	b.	Five teachers T_1, T_2, T_3, T_4 are to be made class teachers for five classes, C_1, C_2, C_3, C_4, C_5, one teacher for each class. T_1 and T_2 do not wish to become the class teachers for C_1 or C_2, T_3 and T_4 for C_4 or C_5, and T_5 for C_3 or C_4 or C_5. In how many ways can the teachers be assigned the work? (Without displeasing any teacher)	**07**	**L3**	**CO4**
	c.	Solve the recurrence relation $F_{n+2} = F_{n+1} + F_n$ where $n \ge 0$ and $F_0 = 0$, $F_1 = 1$.	**07**	**L2**	**CO4**
		Module – 5			
Q.9	**a.**	If G be a set of all non zero real numbers and let $a * b = \frac{ab}{2}$ then show that (G, *) is an abelian group.	**06**	**L2**	**CO5**
	b.	Define Klein group and if A = {e, a, b, c} then show that this is a Klein-4 group.	**07**	**L2**	**CO5**
	c.	State and prove Lagrange's theorem.	**07**	**L2**	**CO5**
		OR			
Q.10	**a.**	If H and K are subgroups of group G, prove that $H \cap K$ is also a subgroup of G. Is $H \cup K$ a subgroup of G?	**06**	**L2**	**CO5**
	b.	Define cyclic group and show that (G, *) whose multiplication table is as given below is cyclic.	**07**	**L2**	**CO5**
	c.	Let $G = S_4$, for $\alpha = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix}$, find the subgroup $H = <\alpha>$. Determine the left cosets of H in G.	**07**	**L3**	**CO5**

*	a	b	c	d	e	f
a	a	b	c	d	e	f
b	b	c	d	e	f	a
c	c	d	e	f	a	b
d	d	e	f	a	b	c
e	e	f	a	b	c	d
f	f	a	b	c	d	e

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Fourth Semester B.E./B.Tech. Degree Supplementary Examination, June/July 2024

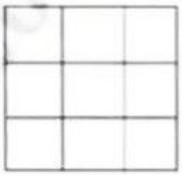
Discrete Mathematical Structures

Time: 3 hrs. Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.

		Module – 1	M	L	C
Q.1	a.	Define Tautology. Show that $[(p \vee q) \wedge (p \rightarrow r) \wedge (q \rightarrow r)] \rightarrow r$ is a tautology by constructing the truth table.	6	L1	CO1
	b.	Prove the following using the laws of logic: $P \rightarrow (q \rightarrow r) \Leftrightarrow (p \wedge q) \rightarrow r$	7	L2	CO1
	c.	Give i) Direct proof ii) indirect proof iii) proof by contradiction for the following statement: "If n is an odd integer then n + 9 is an even integer".	7	L3	CO1
		OR			
Q.2	a.	Test whether the following arguments are valid: $p \rightarrow q$ $r \rightarrow s$ $\underline{\sim q \vee \sim s}$ $\therefore \sim(p \wedge r)$	6	L2	CO1
	b.	Write the following argument in symbolic form and then establish the validity. If a triangle has two equal sides, then it is isosceles. If a triangle is isosceles, then it has two equal angles. The triangle ABC does not have two equal angles. $\therefore$ ABC does not have two equal sides.	7	L1	CO1
	c.	For the following statements, the universe comprises all non-zero integers. Determine the truth value of each statement: i) $\exists x \, \exists y \, [xy = 1]$ ii) $\exists x \, \forall y \, [xy = 1]$ iii) $\forall x \, \exists y \, [xy = 1]$ iv) $\exists x \, \exists y \, [(2x + y = 5) \wedge (x - 3y = -8)]$ v) $\exists x \, \exists y \, [(3x - y = 7) \wedge (2x + 4y = 3)]$	7	L2	CO1
		Module – 2			
Q.3	a.	Prove that $1^2 + 3^2 + 5^2 + \ldots + (2n-1)^2 = \frac{n(2n+1)(2n-1)}{3}$ by mathematical Induction.	6	L2	CO2

	b.	Prove that every positive integer $n \geq 24$ can be written as a sum of 5's and / or 7's.	7	L3	CO2
	c.	Obtain a recursive definition for the sequence $\{a_n\}$ in each of the following cases: i) $a_n = 5n$ ii) $a_n = 3n + 7$ iii) $a_n = 2 - (-1)^n$	7	L3	CO2
		OR			
Q.4	a.	Prove that for any positive integer n, $\sum_{i=1}^{n} \frac{F_{i-1}}{2^i} = 1 - \frac{F_{n+2}}{2^n}$, F_n denote the fibonacci number.	6	L2	CO2
	b.	How many arrangement are there for ail the letters in the word "SOCIOLOGICAL". In how many of these arrangements. i) A and G are adjacent ii) All vowels are adjacent.	7	L2	CO2
	c.	Determine the coefficient of $a^2b^3c^2d^5$ in the expansion of $(a + 2b - 3c + 2d + 5)^{16}$.	7	L2	CO2
		Module – 3			
Q.5	a.	Let A = {1, 2, 3, 4, 6} and R be a relation on A defined by $a^R b$ if and only if "a is a multiple of b". Write down the relation R, relation matrix M(R) and draw its digraph. List out its indegree and out degree.	6	L2	CO3
	b.	Let f and g be functions from R to R defined by $f(x) = ax + b$ and $g(x) = 1 - x + x^2$. If $(gof)(x) = 9x^2 - 9x + 3$ determine a and b.	7	L3	CO3
	c.	State Pigeon hole principle. Show that if n + 1 numbers are chosen from 1 to 2n then atleast one pair add to 2n + 1.	7	L2	CO3
		OR			
Q.6	a.	Let $f : R \to R$ be defined by $f(x) = \begin{cases} 3x - 5, \text{if } x > 0 \\ 1 - 3x, \text{if } x \leq 0 \end{cases}$ find f(-1), f(5/3), $f^{-1}(0)$, $f^{-1}(-3)$, $f^{-1}([-5, 5])$ and $f^{-1}([-6, 5])$.	6	L1	CO3
	b.	Let f, g, h be functions from Z to Z defined by $f(x) = x - 1$, $g(x) = 3x$, $h(x) = \begin{cases} 0, \text{if } x \text{ is even} \\ 1, \text{if } x \text{ is odd} \end{cases}$ Determine (fo(goh)) (x), ((fog)oh)(x) and verify that fo(goh) = (fog)oh.	7	L2	CO3
	c.	Draw the Hasse (POSET) diagram which represents positive divisors of 36.	7	L2	CO3
		Module – 4			
Q.7	a.	In how many ways 5 number of a's, 4 number of b's and 3 number of c's, can be arranged so that all the identical letters are not in a single block.	6	L3	CO4

	b.	Four persons P_1, P_2, P_3, P_4 who arrive late for a dinner party find that only one chair at each of five tables T_1, T_2, T_3, T_4 and T_5 is vacant. P_1 will not sit at T_1 or T_2, P_2 will not sit at T_2, P_3 will not sit at T_3 or T_4 and P_4 will not sit at T_4 or T_5. Find the number of ways they can occupy the vacant chairs.	7	L2	CO4
	c.	Solve the recurrence relation $a_n = na_{n-1}$ where $n \geq 1$ and $a_0 = 1$.	7	L2	CO4
		OR			
Q.8	a.	In how many ways can the 26 letters of the English alphabet be permuted so that none of the patterns CAR, DOG, PUN or BYTE occurs?	6	L2	CO4
	b.	Find the rook polynomial for the 3 * 3 board by using the expansion formula. 	7	L2	CO4
	c.	Solve the recurrence relation $F_{n+2} = F_{n+1} + F_n$ where $n \geq 0$ and $F_0 = 0$, $F_1 = 1$.	7	L2	CO4
		Module – 5			
Q.9	a.	Define Group. Show that fourth roots of unity is an abelian group under $\otimes$.	6	L2	CO5
	b.	Define Klein 4 group. Verify A = {1, 3, 5, 7} is a Klein 4 group under $\otimes_8$.	7	L2	CO5
	c.	State and prove Lagrange's theorem.	7	L2	CO5
		OR			
Q.10	a.	If H, K are subgroups of a group G, prove that $H \cap K$ is also a subgroup of G. Is $H \cup K$ a subgroup of G?	6	L2	CO5
	b.	Define cyclic group and show that (G, •) whose multiplication table is as given below is cyclic.	7	L2	CO5
	c.	Prove that the only left coset of a subgroup H of a group G which is also a subgroup of G is H itself.	7	L2	CO5

*	a	b	c	d	e	f
a	a	b	c	d	e	f
b	b	c	d	e	f	a
c	c	d	e	f	a	b
d	d	e	f	a	b	c
e	e	f	a	b	c	d
f	f	a	b	c	d	e

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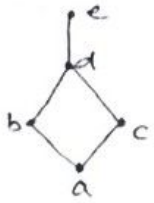
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Fourth Semester B.E./B.Tech. Degree Examination, June/July 2025
Discrete Mathematical Structures

Time: 3 hrs. Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.

		Module – 1	M	L	C
Q.1	a.	Define Tautology, show that $[(p \vee q) \wedge \{(p \to r) \wedge (q \to r)\}] \to r$	6	L1	CO1
	b.	Prove the following using the laws of logic : $\neg [\{(p \vee q) \wedge r\} \to \neg q] \Leftrightarrow \neg [\neg[(p \vee q) \wedge r] \vee \neg q] \Leftrightarrow q \wedge r$.	7	L2	CO1
	c.	Give i) a direct proof ii) an Indirect proof for the following statement "If n is an odd integer then n + 9 is an even integer".	7	L2	CO1
		OR			
Q.2	a.	Define i) an open statement ii) quantifiers.	6	L2	CO1
	b.	Test the validity of the following arguments. i) $\dfrac{p \wedge q}{\underline{p \to (q \to r)}}$ $\therefore r$ ii) $\dfrac{P}{\underline{P \to \sim q}}$ $\dfrac{}{\underline{\sim q \to \sim r}}$ $\therefore \sim r$	7	L2	CO1
	c.	For the following statements the universe comprises all non – zero integers. Determine the truth value of each statement. i) $\exists x, \exists y\ [xy = 1]$ ii) $\exists x, \forall y\ [xy = 1]$ iii) $\forall x, \exists y\ [xy = 1]$ iv) $\exists x, \exists y\ [(2x + y = 5) \wedge (x - 3y = -8)]$ v) $\exists x, \exists y\ [(3x - y = 17) \wedge (2x + 4y = 3)]$.	7	L2	CO1
		Module – 2			
Q.3	a.	Define the well ordering principle. By Mathematical induction, prove that $1 + 2 + 3 + \ldots + n = \frac{1}{2} n(n+1)$, $n \in z^+$.	6	L2	CO2
	b.	Prove that $F_n = \frac{1}{\sqrt{5}} \left[\left(\frac{1+\sqrt{5}}{2} \right)^n - \left(\frac{1-\sqrt{5}}{2} \right)^n \right]$. For F_0 , F_1, F_2 , ... are the Fibonacci numbers.	7	L2	CO2
	c.	Find the number of permutations of the letters of the word 'MASSASAUGA'. In how many of these all four A's are together? How many of them begin with S's?	7	L3	CO2
		OR			

Q.4	a.	Prove that $4n < n^2 - 7$ for all positive integers $n \geq 6$.	6	L2	CO3
	b.	Find the co-efficients of $x^9 y^3$ in the expansion of $(2x - 3y)^{12}$.	7	L3	CO3
	c.	Let $a_0 = 1$, $a_1 = 2$, $a_3 = 3$ and $a_n = a_{n-1} + a_{n-3}$ for $n \geq 3$, prove that $a_n \leq 3^n$ for all +ve integers n.	7	L2	CO3
		Module – 3			
Q.5	a.	State Pigeon hole principle. Prove that if 30 dictionaries in a library contains a total of 61,327 pages then atleast one of dictionaries must have atleast 2045 pages.	6	L2	CO3
	b.	Define power set. For any sets A, B, $C \subseteq U$, prove that $A \times (B \cup C) = (A \times B) \cup (A \times C)$.	7	L2	CO3
	c.	Let f and g be functions from R to R defined by $f(x) = ax + b$ and $g(x) = 1 - x + x^2$ if $(gof)(x) = 9x^2 - 9x + 3$, determine a & b.	7	L3	CO3
		OR			
Q.6	a.	Let $f: R \rightarrow R$ be defined by $f(x) = \begin{cases} 3x - 5, \text{if } x > 0 \\ 1 - 3x, \text{if } x \leq 0 \end{cases}$ Find $f^{-1}(-5, 5)$ and $f^{-1}(-6, 5)$.	6	L2	CO3
	b.	Let N be the set of Natural numbers. Let a relation R be defined by $R = \{(a, b) / a \in N, b \in N, a - b \text{ is divisible by } 5\}$. Prove that R is an equivalence relation.	7	L2	CO3
	c.	For $A = \{a, b, c, d, e\}$, the Hasse diagram for the poset(A, R) is as shown below : i) Determine the relation matrix for R ii) Construct the diagraph for R. 	7	L3	CO3
		Module – 4			
Q.7	a.	Determine the number of integers between 1 and 250 that are divisible by 3 and not divisible by 5 and 7.	6	L3	CO4
	b.	Solve the recurrence relation $F_{n+2} = F_{n+1} + F_n$, where $n \geq 0$ and $F_0 = 0$, $F_1 = 1$.	7	L2	CO4
	c.	Define Derangement. Find the number of derangement of 1, 2, 3, and 4.	7	L3	CO4
		OR			

Q.8	**a.**	Find the Rook polynomial for the chess board contain 4 squares as shown in the Fig.Q8(a). 	1	2
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3	4	 Fig.Q8(a)	**6**	**L3**	**CO4**
	b.	Solve the recurrence relation $a_n = 5a_{n-1} + 6a_{n-2}$, $n \geq 2$, $a_0 = 1$, $a_1 = 3$.	**7**	**L2**	**CO4**
	c.	Find the distinct numbers which are multiples of at least one of 15, 40 and 35 not exceeding 1000.	**7**	**L3**	**CO4**
		Module – 5			
Q.9	**a.**	Define group and subgroup with example each.	**6**	**L1**	**CO5**
	b.	State and prove Lagrange's theorem.	**7**	**L2**	**CO5**
	c.	Define Klein 4 group. Verify A = {e, a, b, c} is a Klein 4 group.	**7**	**L2**	**CO5**
		OR			
Q.10	**a.**	Prove that the intersection of two subgroup of a group is a subgroup of the group.	**6**	**L2**	**CO5**
	b.	Prove that the cube roots of unity form a group under the multiplication.	**7**	**L2**	**CO5**
	c.	Let $G = S_4$, the symmetric group of order 4, for $\alpha = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix}$, find the subgroup $H = \langle a \rangle$, determine the number of left cosets of H in G.	**7**	**L3**	**CO5**

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CBCS SCHEME

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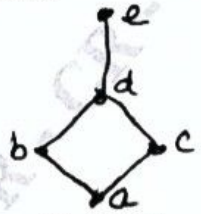
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Fourth Semester B.E./B.Tech. Degree Examination, Dec.2025/Jan.2026
Discrete Mathematical Structures

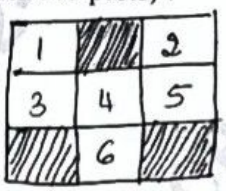
Time: 3 hrs. Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.

		Module – 1	M	L	C
Q.1	a.	Define a logically equivalent. Show that $[(p \vee q) \rightarrow r] \leftrightarrow [\neg r \rightarrow \neg (p \vee q)]$ for primitive statements p, q, r is logically equivalence.	6	L1	CO1
	b.	Establish the validity of the following argument using the rules of inference: $\{(p \rightarrow q) \wedge (r \rightarrow s) \wedge (\neg q \vee \neg s)\} \rightarrow \neg (p \wedge r)$	7	L2	CO1
	c.	i) Give an indirect proof of the statement: "The product of two even integers is an even integer". ii) Provide a proof by contradiction of the following statement: "For every integer n, if n^2 is odd, then n is odd".	7	L3	CO1
		OR			
Q.2	a.	Define a tautology. Prove that, for an propositions p, q, r the compound proposition. $\{p \rightarrow (q \rightarrow r)\} \rightarrow \{(p \rightarrow q) \rightarrow (p \rightarrow r)\}$ is tautology.	6	L1	CO1
	b.	Using law of logic, prove the result. $[(p \vee q) \wedge (p \vee \neg q)] \vee q \Leftrightarrow p \vee q$	7	L2	CO1
	c.	Find whether the following argument is valid: If a triangle has two equal sides, then it is isoceles. If a triangle is isoceles, then it has two equal angles. A certain triangle PQR does not have two equal angles. $\therefore$ The triangle PQR does not have two equal sides.	7	L3	CO1
		Module – 2			
Q.3	a.	By mathematical induction, prove that $(n!) \geq 2^{n-1}$ for all integers $n \geq 1$.	6	L2	CO2
	b.	Find an in explicit form for the following: i) $a_1 = 4, a_n = a_{n-1} + n$ for $n \geq 2$ ii) $a_1 = 7, a_n = 2a_{n-1} + n$ for $n \geq 2$	7	L2	CO2
	c.	Find how many distinct four-digit integers one can make from the digits 1, 3, 3, 7, 7, 8	7	L3	CO2

		OR			
Q.4	**a.**	Prove that $4n < (n^2 - 7)$ for all positive integers $n \geq 6$.	**6**	**L2**	**CO2**
	b.	i) How many arrangements are there for all letters in the word SOCIOLOGICAL? ii) In how many of these arrangements all the vowels are adjacent?	**7**	**L3**	**CO2**
	c.	Determine the coefficient of i) xyz^2 in the expansion of $(2x - y - z)^4$ ii) $x^2y^2z^3$ in the expansion of $(3x - 2y - 4z)^7$	**7**	**L2**	**CO2**
		Module – 3			
Q.5	**a.**	Let A = {1, 2, 3, 4, 5, 6} and B = {6, 7, 8, 9, 10}. If a function f : A → B is defined by f = {(1, 7), (2, 7), (3, 8), (4, 6), (5, 9), (6, 9)} determine $f^{-1}(6)$ and $f^{-1}(9)$. If B_1 = {7, 8} and B_2 = {8, 9, 10}, find $f^{-1}(B_1)$ and $f^{-1}(B_2)$.	**6**	**L2**	**CO2**
	b.	State Pigeon-hole principle. ABC is an equilateral triangle whose sides are of length 1cm each. If we select 5 points inside the triangle, prove that atleast two of these points are such the distance between them is less than 1/2 cm.	**7**	**L3**	**CO2**
	c.	Let f and g be functions from R to R defined by f(x) = ax + b and $g(x) = 1 - x + x^2$ if (gof) (x) = $9x^2 - 9x + 3$, determine a & b.	**7**	**L2**	**CO2**
		OR			
Q.6	**a.**	Let f, g, h be functions from z to z defined by f(x) = x – 1, g(x) = 3x, $h(x) = \begin{cases} 0, & \text{if } x \text{ is even} \\ 1, & \text{if } x \text{ is odd} \end{cases}$ Determine (f o (g o h)) (x) and ((f o g) o h) (x) and verify that f o (g o h) = (f o g) o h.	**6**	**L2**	**CO3**
	b.	For A = {a, b, c, d, e}, the Hasse diagram for the poset (A, R) is as shown in Fig.Q.6(b). i) Determine the relation matrix for R. ii) Construct the diagraph for R.  Fig.Q.6(b)	**7**	**L3**	**CO3**
	c.	Let A = {1, 2, 3, 4} and let R be the relation on A defined by xRy if and only if "x divides y", i) Write down R as a set of ordered pairs ii) Draw the diagraph of R iii) Determine the in-degrees and out-degrees of the vertices in the digraph.	**7**	**L3**	**CO3**

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		Module – 4			
Q.7	**a.**	In how many ways 5 number of a's, 4 number of b's and 3 number of c's can be arranged so that all the identical letters are not in a single block.	**6**	**L2**	**CO4**
	b.	Find the number of permutations of the letters a, b, c, ... x, y, z in which none of the patterns spin, game, path or net occurs.	**7**	**L3**	**CO4**
	c.	Solve the recurrence relation $a_n - 6a_{n-1} + 9a_{n-2} = 0$ for $n \geq 2$, given that $a_0 = 5$, $a_1 = 12$.	**7**	**L2**	**CO4**
		OR			
Q.8	**a.**	Find the number of derangements of 1, 2, 3, 4. Mention them.	**6**	**L2**	**CO4**
	b.	By using the expansion formula, find the rook polynomial for the board c shown below (madeup of unshaded parts) : 	**7**	**L3**	**CO4**
	c.	Solve the recurrence relation $a_n - 3a_{n-1} = 5 \times 3^n$ for $n \geq 1$ given that $a_0 = 2$	**7**	**L2**	**CO4**
		Module – 5			
Q.9	**a.**	Let $w_4 = \{1, -1, i, -i\}$, the set of all fourth roots of unity. Show that w_4 is an abelian group under x.	**6**	**L2**	**CO5**
	b.	Define Klein-4-group. If A = {e, a, b, c} then show that this is a Klein-4 group.	**7**	**L2**	**CO5**
	c.	State and prove Lagrange's theorem.	**7**	**L2**	**CO5**
		OR			
Q.10	**a.**	Show that the group (G, *) whose multiplication table is as given below is cyclic. * | a b c d e f a | a b c d e f b | b c d e f a c | c d e f a b d | d e f a b c e | e f a b c d f | f a b c d e CMRIT LIBRARY BANGALORE - 560 037	**6**	**L**	**CO**
	b.	Let $G = S_4$, the symmetric group of degree 4. For $\alpha = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix}$. Find the subgroup $H = \langle \alpha \rangle$. Also, determine the number of left cosets of H in G.	**7**	**L3**	**CO5**
	c.	If * is an operation on z defined by $x * y = x + y + 1$, prove that (z, *) is an abelian group.	**7**	**L2**	**CO5**

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